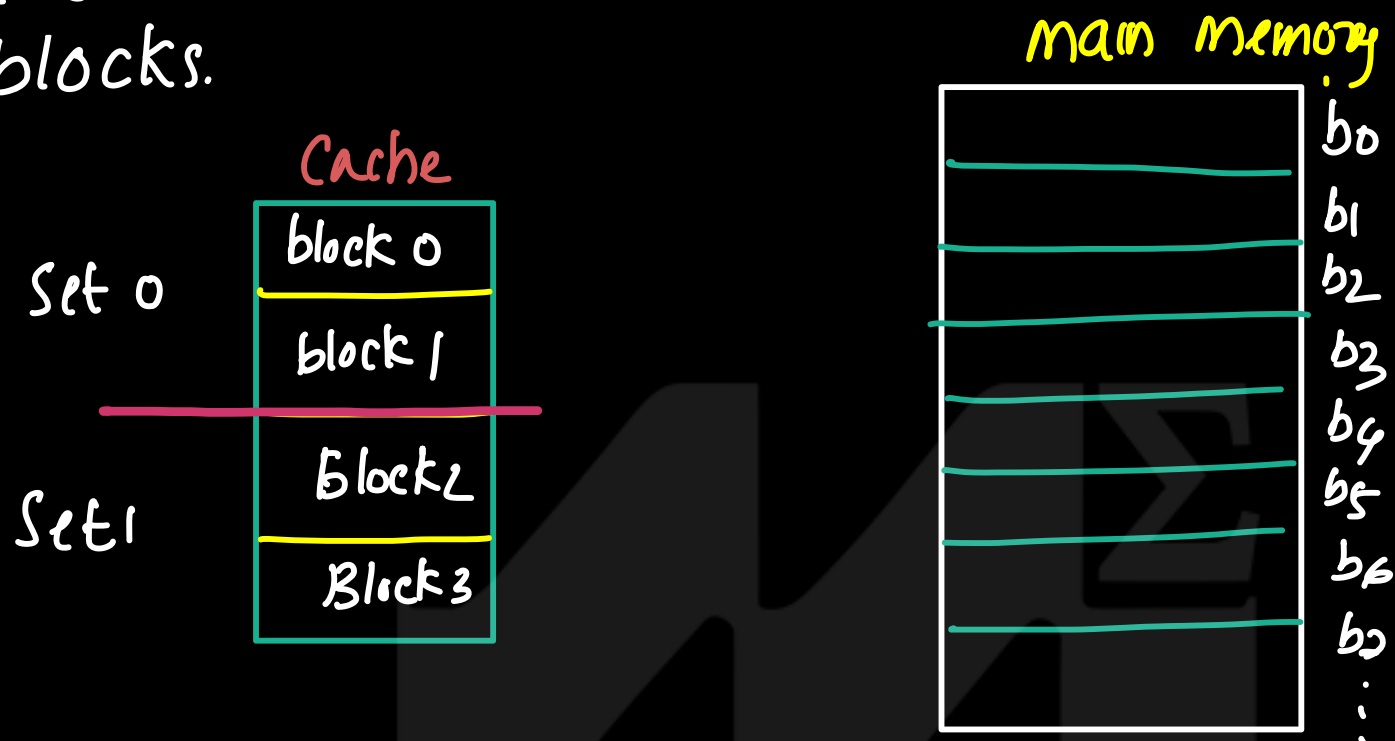


Set associative Cache Mapping

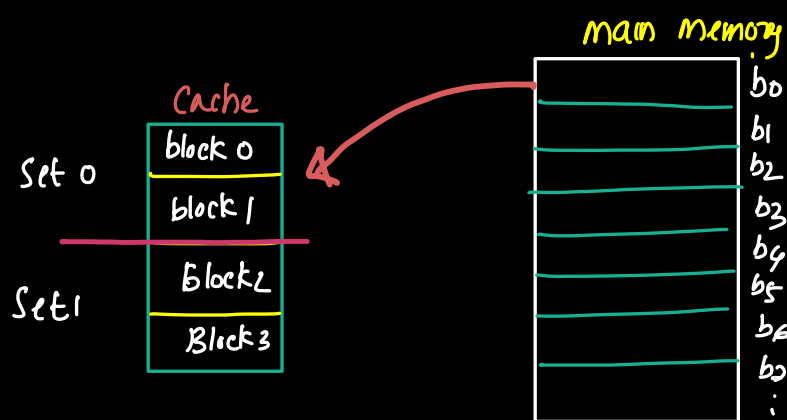
Note

- Explain cache mapping (Refer previous Note)
- The major problem of direct mapping is conflict miss.
- This problem can be overcome using set associative mapping
- In set associative mapping, the cache is organised into several sets. and each set contains 2 or more blocks.



- Each block in the main memory is mapped to a particular set and memory block can be placed in any block within the particular set.

eg: Memory block '0' is mapped to set '0'. In set 0 there are two blocks. so Memory block b0 can be mapped any block within the set 0.



- This freedom helps to overcome conflict miss.
- If N blocks are present in a cache set. then

It is called N-associative cache

(N)

eg. If two blocks are present in a set, then it is called two way set associative mapping.

Two-way set-associative Mapping

— Consider a main memory with following specification

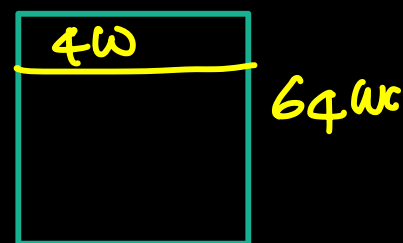
$$\text{MM Size} = 64 \text{ Byte}$$

$$1 \text{ Byte} = 1 \text{ word}$$

$$\text{MM Size} = 64 \text{ word}$$

$$\text{MM Block size} = 4 \text{ Byte} = 4 \text{ word}$$

$$\text{no. of Block} = \frac{64}{4} = 16 \text{ block.}$$



$$\text{Cache size} = 16 \text{ Byte} = 16 \text{ word}$$

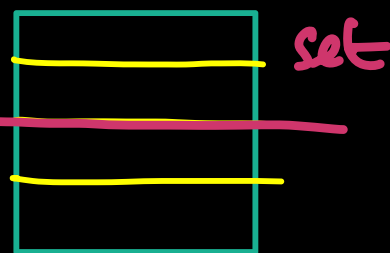
$$\text{Cache Block size} = \text{main Block size} = 4 \text{ B}$$

$$\text{Total cache Block} = \frac{\text{CS}}{\text{BS}} = \frac{16}{4} = \underline{\underline{4}}$$

$$\text{No. of Block in set} = 2$$

$$\text{Total No. of Set} = \frac{\text{Total No. of Block}}{\text{No. of Block in set}}$$

$$= \frac{4}{2} = \underline{\underline{2}}$$



Set No	block no	Cache Memory (16 bytes)
0	0	
	1	
1	2	
	3	

Block No	Main Memory (64 bytes)			
0				
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				

— Memory block 0 can be mapped to any block in the set 0.

mapping function

$$\text{Set Number} = (\text{memory block no}) \bmod \text{no. of set}$$

eg: block 15 is mapped to set no 1
 $\text{Set Num} = 15 \% 2 = 1 \rightarrow \text{set no: 1} \text{ (2)}$

→ Address field format for set associative mapping.
 A physical Address is divided into 3 field

tag bits	Set bits	Block offset
----------	----------	--------------

- ① Block offset field
- ② Set bit field
- ③ tag bits.

- Set bit field

these bits are used to identify which set no should be accessed

→ tag bit field

these bits are used to identify the which Block in the main memory is stored in the selected set

→ Block offset

these bits are used to identify the which word inside the blocks should be needed.

physical Address

tag bits.	Set bit	Byte offset
-----------	---------	-------------

Eg: — Consider a main memory with following specification

MM Size = 64 Byte
 2 Byte = 1 word

— physical Address (PA)
 no. of bits in PA = $\log_2 \text{MM Size} = \log_2 64 = \underline{6} \text{ bit}$



— Block offset

$$\text{Block size} = 4 \text{ Byte} = 4 \text{ word}$$

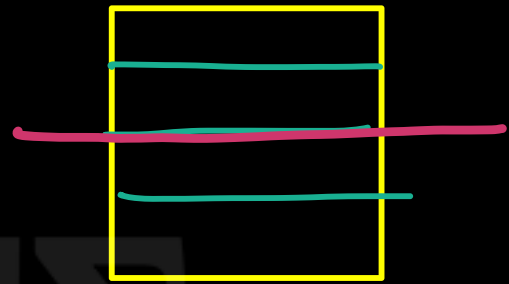
$$\text{Block offset} = \log_2 BS = \log_2 4 = \underline{2 \text{ bit}}$$



→ Set bit

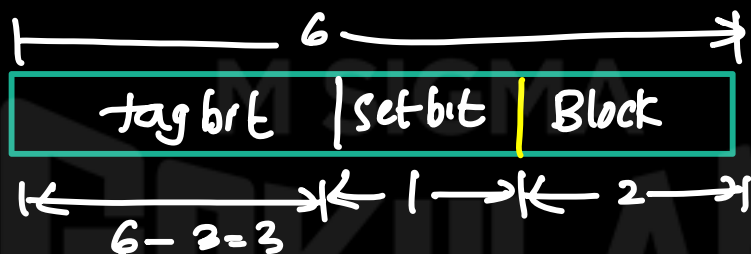
Total No. Set in cache = ?

$$\begin{aligned} \text{Total no. of block} &= \frac{CS}{\text{Block size}} \\ &= \frac{16B}{4B} = \underline{4} \end{aligned}$$



$$\text{Total No. of set} = \frac{\text{Total no. of Block}}{\text{no. of Block in set}} = \frac{4}{2} = \underline{2}$$

$$\text{Set bit} = \log_2 2 = 1 \text{ bit}$$



How the CPU Accesses Data from Cache(2 way set associative)

Note

Processor generates a physical address (PA)

When the CPU needs data or an instruction, it generates a physical address.

This physical address is divided into three fields:

- Tag
- Set index (set bits)
- Block/offset

Cache organization into sets

- The cache is divided into multiple sets.
- Each set contains more than one block (let say 2)-
- Each set contains two ways
- Each way consists
 - Tag field - identifies which block of main memory is stored
 - Valid bit - indicates whether the data in that line is valid
 - Data field - actual data copied from main memory

Selecting the set using set bits

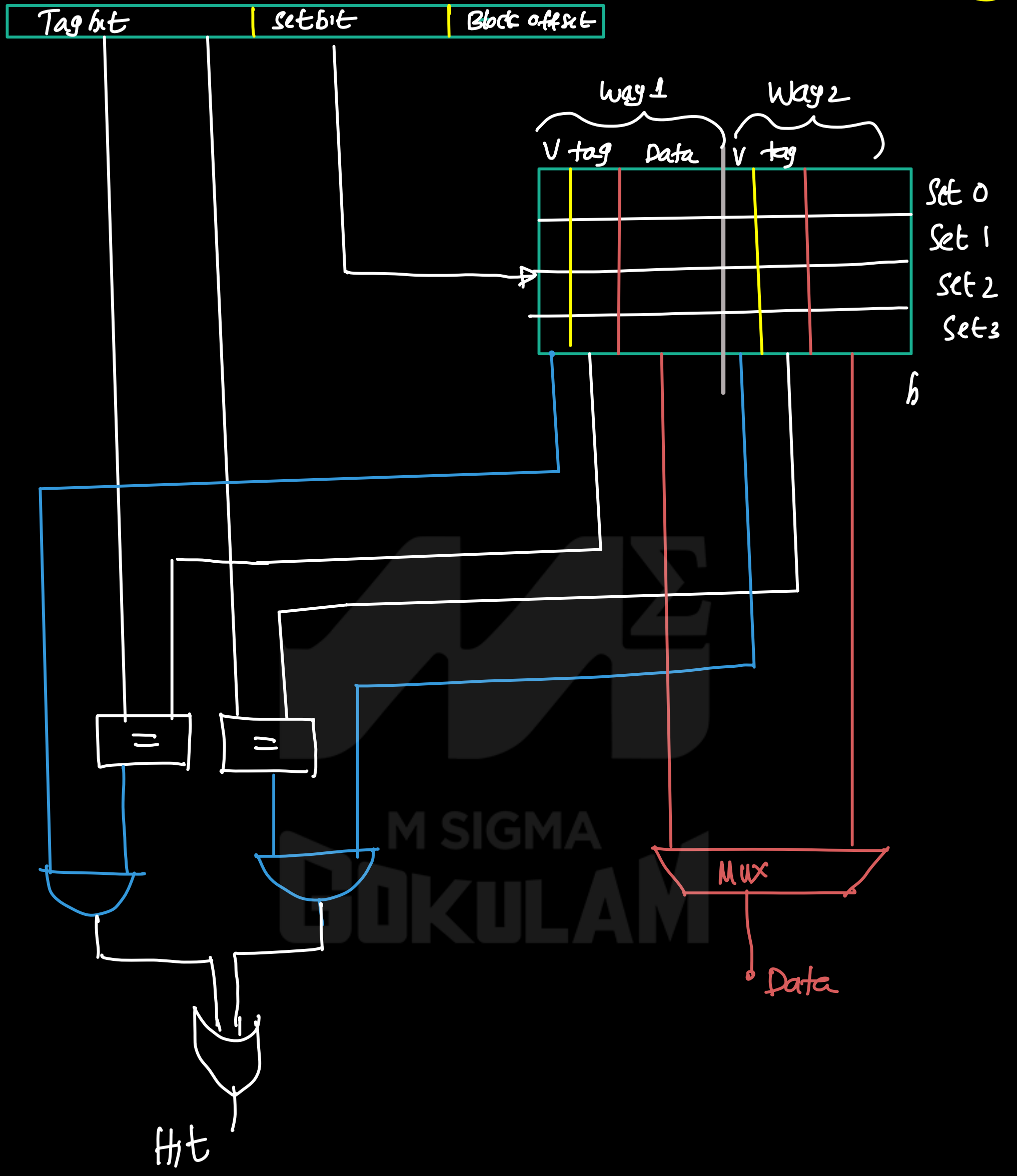
- The CPU uses the set bits from the physical address to select one specific set in the cache.
- These set bits determine which set may contain the required block of main memory.

Tag comparison

- Within the selected set, the tag field from the physical address is compared with the tag stored in the each way(2 blocks).
- This comparison checks whether the required memory block is present in any one of the way
- If the tag bits match, then check for the valid bit for the corresponding block in the selected set . if The valid bit is 1, then required data is available (cache hit)

Data access

- When a cache hit occurs, the CPU directly copies the required data from the data field of the selected set through a multiplexer.



Advantages of Set Associative Mapping

Reduces conflict misses

Since a memory block can be placed in any block within a set, conflict misses are much lower than in direct mapping.

Better cache hit ratio

Allowing multiple blocks per set increases the chance that required data is already present in the cache.

More flexible than direct mapping

Unlike direct mapping, blocks are not restricted to a single cache location.

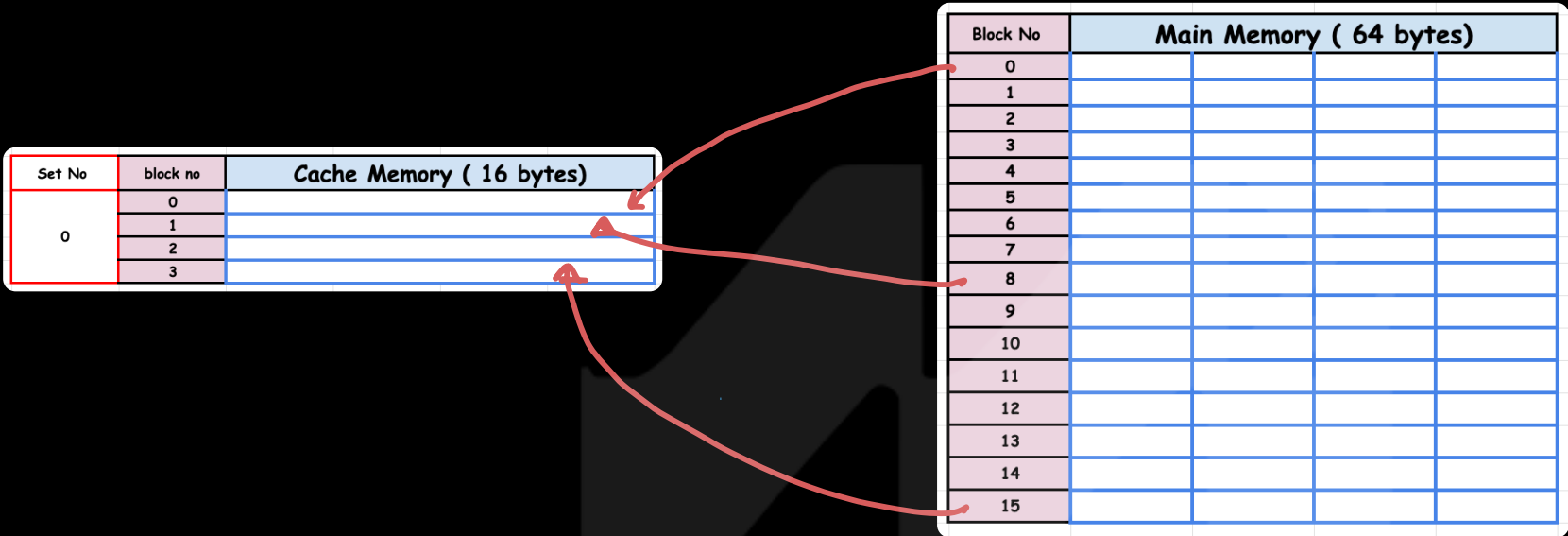


Fully Associative Mapping

- A fully associative cache contains a single set with multiple blocks

Set No	block no	Cache Memory (16 bytes)
0	0	
	1	
	2	
	3	

- Fully associative mapping is a cache mapping technique in which any block of main memory can be stored in any block of the cache. There is no fixed mapping between main memory blocks and cache blocks.



Address Format

In fully associative mapping, the physical address generated by the CPU is divided into:

Tag
Block offset

Address bits	
Tag bits	Block Offset

Cache Organisation



Each cache block contains:

Tag field – identifies the main memory block stored

Valid bit – indicates whether the data is valid

Data field – stores the actual data

Working of Fully Associative Mapping

The CPU generates a physical address.

The tag field of the address is compared with the tags of all cache blocks simultaneously.

If a matching tag is found and the valid bit is 1, a cache hit occurs.

The required data is read from the corresponding cache block.

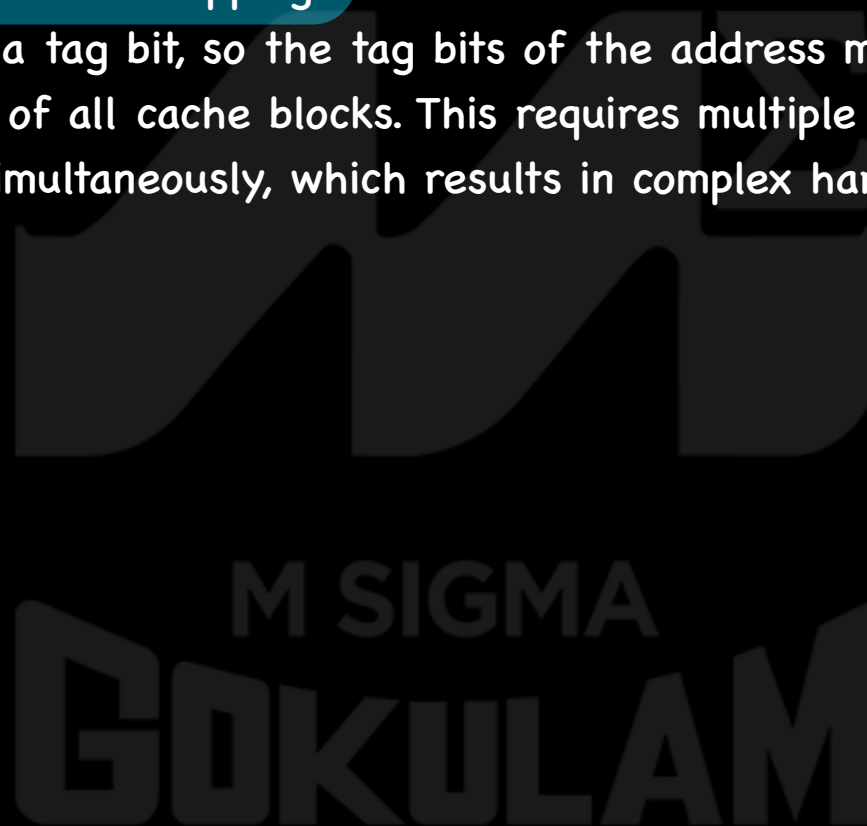
Advantages of Fully Associative Mapping

- Eliminates conflict misses

- Highest cache hit ratio

Disadvantages of Fully Associative Mapping

Each block in a set has a tag bit, so the tag bits of the address must be compared with the tag bits of all cache blocks. This requires multiple comparators to compare all the blocks simultaneously, which results in complex hardware.



Comparison of Cache Mapping Techniques			
Feature	Direct Mapping	Set Associative Mapping	Fully Associative Mapping
Block placement	Each main memory block maps to one specific cache block	Each main memory block maps to one set and can be placed in any block within that set	A main memory block can be placed in any cache block
Cache organisation	Cache divided into Sets (1 block per set)	Cache divided into sets, each set has 2 or more blocks	Cache consists one set with many block
Address fields	Tag, set bit, block Offset	Tag, set bit, block Offset	Tag, Block Offset
Conflict miss	High	Low	None
Cache hit ratio	Low	Medium	High
Hardware complexity	Very simple	Moderate	Very complex
Cost	Low	Medium	High
Access time	Fast	Moderate	Slower

Summary

- Direct mapping is simple and fast but suffers from high conflict misses.
- Set associative mapping reduces conflict misses and offers a balance between cost and performance.
- Fully associative mapping eliminates conflict misses completely but requires complex and expensive hardware.

- Explain the memory system architecture and memory hierarchy in detail.
- Explain cache memory and its basic concepts with a neat diagram.
- Describe direct mapping cache with address format and example.
- Explain set associative cache mapping using a 2-way set associative cache with a numerical example.
- Explain fully associative mapping with advantages and disadvantages.
- Compare all cache mapping techniques with suitable diagrams.
- Explain cache performance analysis and average access time.
- Discuss the problems of direct mapping and how they are solved using set associative mapping.
- A system has a main memory and cache. Given suitable parameters, calculate hit ratio and access time. (Numerical)

